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Post-Quantum AI Provenance Watermarking

Theme-Quantum Security

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EL Phase-I
Presentation Outline

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Introduction

Post-Quantum AI Provenance Watermarking,is a framework designed to secure AI training datasets and models against data manipulation and future quantum computing threats.

Key Details

- **Importance of Data Quality in AI:**
  - AI is increasingly used in critical sectors like healthcare, education, and finance.
  - The reliability and accuracy of AI decisions depend directly on training data quality.
- **Primary Challenges:**
  - **Data Manipulation:** Training datasets can be modified, corrupted, or swapped without easy detection.
  - **Quantum Threat:** Advancements in quantum computing threaten traditional cryptographic security.
- **Proposed Solution — *Post-Quantum AI Provenance Watermarking*:**
  - Integrates **Post-Quantum Cryptography**, **Dataset Provenance**, and **AI Model Watermarking**.
- **Core Functions & Objectives:**
  - Tracks the origin and verifies the integrity of training datasets.
  - Establishes a verifiable link between training data and trained AI models.
  - Enhances **transparency**, **traceability**, and **trust** across security-critical AI systems.

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Literature Review

1. Poisoned Acoustics: Targeted Data Poisoning Attacks on Acoustic Vehicle Classification at Sub-1% Corruption Rates **Author**- Harrison Dahme(CTO and Partner, Hack VC) **Year:** 2026 **Source**- https://arxiv.org/pdf/2602.22258	**Key Findings:** • Label-flipping attacks achieved a 95.7% success rate with only 0.5% corrupted data, remaining undetected by overall model accuracy. • Proposes content-addressed hashing, Merkle trees, and ML-DSA-65 signatures to enable verifiable dataset provenance. **Limitations:** • Cannot detect if data was maliciously poisoned prior to signing, and does not directly address AI model watermarking.
2. AI Supply Chain Security: MBOM-PQC Provenance, PQC Attestation, and a Maturity Model for Quantum-Resistant Assurance **Author:** Robert Campbell **Year:** 2026 **Journal:** *Systems, 14(5), 593*	**Key Findings:** • Explores PQC-based provenance and attestation mechanisms. **Limitations:** • Focuses primarily on architecture and security maturity without demonstrating a specific watermarking technique to reliably preserve dataset provenance.

3. Robust Watermarking for Diffusion Models Using Error-Correcting Codes and Post-Quantum Key Encapsulation **Authors:** Xianglei Hu, Beining Zhang, Mawaheb Al-Dossari, N. S. Abd El-Gawaad, Mira Rakhimzhanova, and Ahmad Saeed Khan **Year:** 2026 **Journal:** *Frontiers in Physics*	**Key Findings:** • Uses error-correcting codes and Kyber-based post-quantum key encapsulation to secure watermark key exchanges and improve recovery after content modification. • Evaluates watermarking robustness against various forms of image manipulation. **Limitations:** • Focuses specifically on diffusion-generated images rather than dataset provenance or AI model weight watermarking.
4. Enhancing Quantum Security over Federated Learning via Post-Quantum Cryptography **Authors**: Pingzhi Li,Tianlong Chen & Junyu Liu **Year**: 2024 **Source**: https://arxiv.org/pdf/2409.04637	**Key Findings:** • Investigates the use of post-quantum cryptographic techniques in federated-learning environments. • Addresses security challenges associated with communication between distributed participants. **Limitations:** • The research focuses on securing federated-learning environments rather than directly watermarking model parameters.

5. Post-Quantum Verifiable Joint-Watermarking for Multiparty Artificial Intelligence Models **Author**: Vipin Singh Sehrawat, Fremont **Year**: 2026 **Source**: https://image-ppubs.uspto.gov/	**Key Findings:** • Investigates joint watermarking and cryptographic verification for AI models developed by multiple contributors. • Solves the challenge of contribution attribution and ownership tracking in collaborative AI environments. **Limitations:** • Does not directly address dataset-level provenance or training-data integrity.
6. Quantum-Inspired Post-Quantum Secure DCT Watermarking for Tamper Detection and Localization in Medical Images	**Key Findings:** • Focuses on detecting and identifying the location of image tampering. • Explores the integration of quantum-inspired methods with post-quantum security concepts. **Limitations:** • Focuses only on individual medical images rather than securing entire AI training datasets or establishing model training history.

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Problem Definition

The Vulnerability (Data Poisoning)

- AI models are essentially "black boxes." If malicious data is injected during training, the model silently learns dangerous behaviors that are nearly impossible to detect post-deployment.

The Quantum Timebomb (Q-Day)

- We currently rely on traditional digital signatures (RSA/ECC) to authenticate datasets. Within the decade, quantum computers will easily break these cryptographic standards.

The Core Problem

- Once traditional encryption falls, attackers can mathematically forge signatures on poisoned datasets. Currently, there is no future-proof infrastructure to guarantee an enterprise AI model's training data is authentic, unaltered, and safe from quantum-era tampering.

The Bottom Line: *We lack a verifiable, quantum-resistant "digital birth certificate" for high-stakes AI models.*

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Objectives

Main Objective:

To develop a post-quantum-secure AI provenance system that verifies the authenticity of training data, tracks the origin of AI models, and improves trust in the AI development process.

Specific Objectives:

- To implement post-quantum cryptographic techniques, such as ML-DSA, for digitally signing and verifying AI training datasets and provenance records.
- To create a secure system for tracking the connection between training data, training processes, and trained AI models.
- To integrate AI watermarking techniques to help identify model ownership and detect unauthorized model use or modification.
- To evaluate the system's ability to detect data tampering, verify provenance, and maintain security against potential future quantum-based attacks.

Expected Contributions to the Field

- A framework that combines post-quantum cryptography, data provenance, and AI watermarking into a unified system.
- Improved traceability and accountability throughout the AI model development lifecycle.
- A practical approach for detecting unauthorized changes to datasets and verifying the origin of AI models.
- Experimental results showing the strengths and limitations of post-quantum-secure AI provenance techniques.
- A foundation for developing more transparent, secure, and trustworthy AI supply chains.

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Methodology

Design:

The project will follow a **quantitative and experimental research design**. It will involve developing a prototype that combines post-quantum cryptography, AI provenance tracking, and model watermarking. Experiments will be conducted to evaluate the system's security, reliability, performance, and ability to detect tampering.

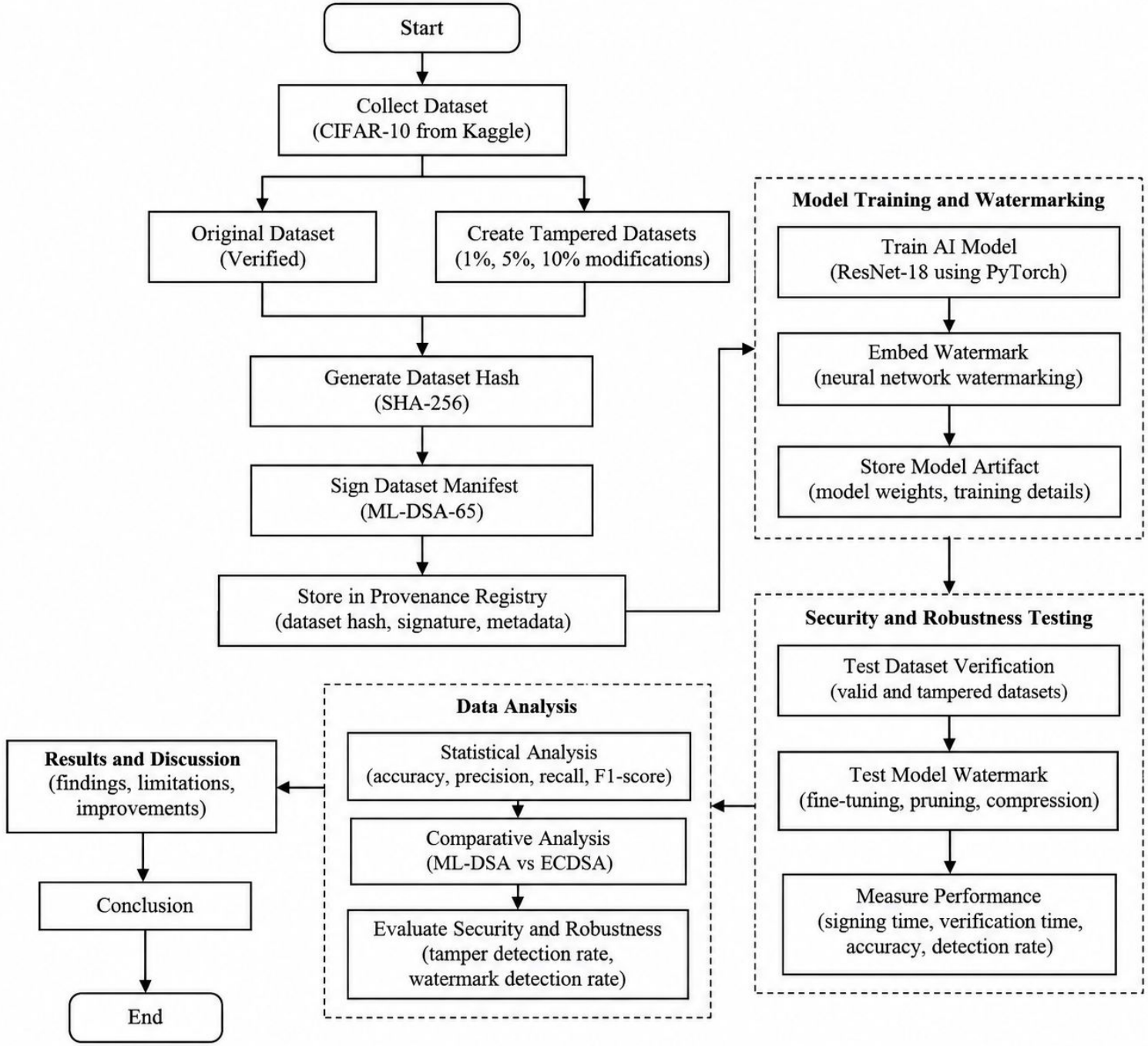
Data Collection Methods:

- **Dataset Collection:** We will use the CIFAR-10 dataset from Kaggle for training and testing.
- **Tampering Experiments:** Create modified datasets with 1%, 5%, and 10% data alterations.
- **Model Training:** Train a ResNet-18 model using both original and tampered datasets.
- **Cryptographic Testing:** Record ML-DSA-65 signing and verification times.
- **Watermarking Experiments:** Test model watermarks after fine-tuning, pruning, and compression.
- **System Logs:** Collect dataset hashes, signatures, model details, and verification results.

Methodology

Data Analysis Techniques

- **Tamper Detection Analysis:** Compare dataset hashes to identify unauthorized modifications.
- **Performance Analysis:** Measure ML-DSA-65 signing time, verification time, and signature size.
- **Model Evaluation:** Compare accuracy, precision, recall, and F1-score for models trained on original and tampered datasets.
- **Watermark Analysis:** Calculate watermark detection rates after fine-tuning, pruning, and compression.
- **Comparative Analysis:** Compare ML-DSA-65 with ECDSA in terms of speed and computational overhead.
- **Statistical Analysis:** Calculate averages and standard deviations from repeated experiments.


Start
Collect Dataset
(CIFAR-10 from Kaggle)
Original Dataset
(Verified)
Create Tampered Datasets
(1%, 5%, 10% modifications)
Generate Dataset Hash
(SHA-256)
Sign Dataset Manifest
(ML-DSA-65)
Store in Provenance Registry
(dataset hash, signature, metadata)
Model Training and Watermarking
Train AI Model
(ResNet-18 using PyTorch)
Embed Watermark
(neural network watermarking)
Store Model Artifact
(model weights, training details)
Security and Robustness Testing
Test Dataset Verification
(valid and tampered datasets)
Test Model Watermark
(fine-tuning, pruning, compression)
Measure Performance
(signing time, verification time,
accuracy, detection rate)
Data Analysis
Statistical Analysis
(accuracy, precision, recall, F1-score)
Comparative Analysis
(ML-DSA vs ECDSA)
Evaluate Security and Robustness
(tamper detection rate,
watermark detection rate)
Results and Discussion
(findings, limitations,
improvements)
Conclusion
End

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Timeline of the project

Initiation Phase (Weeks 1–2)

- Finalize the project objectives and scope.
- Study AI provenance, watermarking, and post-quantum cryptography.
- Identify datasets, tools, and technical requirements.
- Conduct a feasibility study.

Planning Phase (Weeks 3–4)

- Select the CIFAR-10 dataset and ResNet-18 model.
- Design the provenance verification and watermarking workflow.
- Assign tasks and responsibilities among team members.
- Prepare the experimental plan and project schedule.

Execution Phase (Weeks 5–10)

- Collect and prepare the dataset.
- Implement SHA-256 hashing and ML-DSA-65 signatures.
- Develop the provenance tracking system.
- Train the AI model and embed the watermark.
- Conduct tampering and robustness experiments.
- Analyze and document the results.

Closure Phase (Weeks 11–13)

- Complete system testing and final documentation.
- Evaluate the results and identify limitations.
- Prepare the project report and presentation.
- Present the project for review and incorporate feedback.
- Finalize the project and document future improvements.

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Expected Outcomes

1. Expected Results

- Development of a prototype that verifies the authenticity of AI training datasets using **SHA-256 hashing and ML-DSA-65 digital signatures**.
- Successful detection of unauthorized dataset modifications and tampering.
- Creation of a provenance record linking the dataset, training process, and trained AI model.
- Evaluation of model watermarking for identifying model ownership and detecting unauthorized modifications.
- Measurement of cryptographic performance, model accuracy, verification time, and watermark detection rate.

2. Potential Impact on the Field

- **Unlocking Regulated Industries:** Removes the liability barrier for deploying AI in defense, healthcare, and finance by offering mathematical proof of data integrity.
- **Driving Global Standards:** Positions this architecture to become a foundational requirement for upcoming regulatory frameworks (like the EU AI Act) and future NIST AI safety guidelines.
- **A Paradigm Shift in AI Security:** Moves the industry away from reactive defenses (trying to filter bad outputs) to proactive defense (embedding immutable trust directly into the model's structural weights).
- **Future-Proofing the AI Supply Chain:** Neutralizes the existential "Q-Day" threat, ensuring that multi-million-dollar enterprise AI models remain secure, auditable, and legally defensible for decades.

3. Future Research Directions

- Integration of blockchain or distributed ledgers for decentralized provenance management.
- Development of stronger watermarking techniques for different AI architectures, including large language models.
- Testing the system against advanced data-poisoning and model-tampering attacks.
- Integration with cloud-based MLOps platforms for real-time provenance monitoring.
- Investigation of other post-quantum algorithms and optimization techniques to reduce computational overhead.

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Conclusion

- This project proposes a post-quantum cryptographic framework that safeguards the AI supply chain by embedding provenance watermarks directly into training data and model weights. It combines three pillars — data provenance, deep model watermarking, and post-quantum cryptography (ML-DSA) — to ensure that AI systems used in critical domains like healthcare and education cannot be silently corrupted through data poisoning, even by quantum-enabled attackers.
- As AI becomes deeply embedded in high-stakes decision-making, trust in the data behind these models is no longer optional — it is foundational. By addressing a threat that most current AI security efforts overlook (long-term cryptographic integrity), this project takes a proactive step toward building AI systems that are secure not just today, but for a post-quantum future.

Call to Action or Next Steps:

- Develop a working prototype implementing ML-DSA-based data signing and watermark embedding on a small-scale model.
- Benchmark the system's performance overhead against traditional RSA/ECC-based provenance methods.
- Extend the framework toward standardization, positioning it as a candidate for future "AI provenance certification" requirements in regulated industries.

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Expected Outcome by the end of Second Year

Conference Publication

- Prepare a research paper based on the proposed system and experimental results.
- Describe the implementation of ML-DSA-65, dataset provenance tracking, and AI model watermarking.
- Present the results of dataset-tampering detection, cryptographic verification, and watermark robustness testing.
- Discuss the limitations, practical applications, and future improvements of the system.
- Format the paper according to the **IEEE research paper guidelines**.

Possible Future Outcome

If the prototype demonstrates significant novelty and practical value, the project may be extended into a **research proposal** for developing a larger, cloud-based AI security and provenance platform

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Thank You!